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Tools

- Wire strippers, cutters, and crimping multi-tools
- Digital multimeters, 6000-count true RMS
- Test hook clip lead sets and alligator-clip test leads
- USB logic analyzers (8-channel, 24 MHz)
- Soldering iron stations (60 W) with tips, solder, solder sucker, and tweezers (Restricted — used only at the designated hot-work bench)
- Solder fume extractors (Restricted — hot-work bench)
- Hot glue guns and glue sticks, full-size and mini (Restricted — hot-work bench)
- Clear anti-fog safety glasses (personal protective equipment)
- Soldering practice kits (6-digit alarm clock)

Fasteners, Mechanical, and Enclosure Materials

- M3 screw assortment (750 pcs, 6–30 mm)
- M3 heat-set inserts with insert tip
- Cable/zip ties, assorted 4–12 in.
- Velcro cinch straps and heavy-duty hook-and-loop mounting strips
- Electrical tape, gaffer tape, duct tape, aluminum foil HVAC tape
- Conductive copper foil tape
- Foam board / polystyrene signboard and corrugated cardboard sheets
- Neodymium magnet assortments (Restricted)
- GT2 timing belts and 2GT 36-tooth pulleys
- Propellers
- Binder clips, rubber bands, and white school glue

Electrical Components

- Micro-USB, mini-USB, USB-A, USB-B, and USB-C cables
- USB-C to USB-A cables and adapters
- USB-C breakout boards and USB-C panel-mount crimp jacks
- DC barrel jack power connectors
- Rocker switches, 30 A DC power switches, tactile push buttons, arcade pushbuttons, and micro limit switches with roller lever
- Potentiometers

- 5 V relay modules
- Automotive blade fuse assortment kits (5–30 A)
- MOSFET PWM switch modules
- INA219 current/voltage monitor breakouts
- WAGO 221 lever-nut splicing connectors
- Break-away header pin strips
- 30 AWG enameled magnet wire

Power, Batteries, and Distribution

- Single-cell 3.7 V lithium-polymer packs, 1,200 and 2,000 mAh, JST connector (Restricted)
- 3S 11.1 V lithium-polymer packs — 2,200 mAh XT60 and 5,200 mAh EC5 (Restricted)
- 12 V lithium-ion packs, 2,400 and 5,200 mAh (Restricted)
- USB power banks, 10,000 mAh 30 W (Restricted)
- 9 V and AA alkaline batteries
- TP4056 USB-C and micro-USB lithium charge/protection modules (Restricted)
- XT60 connector pairs with heat-shrink tubing
- 5 V power supplies — 2 A micro-USB, 4 A barrel jack — and 27 W USB-C Raspberry Pi 5

supplies

- LM2596 DC-DC buck converter modules with voltmeter display
- Surge-protected power strips with 6 ft cords (deployed by HackMIT staff only)

Display Components

- Small and large display modules (Elegoo kit stock)
- 1602 I2C character LCD modules
- 0.91 in., 0.96 in., and 1.3 in. monochrome OLED breakouts (SSD1306, I2C)
- 84×48 monochrome LCD modules (Nokia 5110 compatible)
- 2.8 in. TFT touchscreen with SD card socket
- Mini displays for Raspberry Pi
- 7 in. HDMI IPS touchscreen monitors for Raspberry Pi (1024×600)
- Portable 15.6 in. 1080p monitors
- RGB full-color LED matrix panel, 4 mm pitch, 64×32, and LED arrays
- WS2812B addressable LED strips and rings, RGB LED SMD modules, and weatherproof

NeoPixel strip

- HDMI, micro-HDMI, mini-HDMI, DisplayPort, DVI-D, and VGA cables; active

DisplayPort-to-HDMI and HDMI-to-USB-C adapters

Microcontrollers and Development Boards

- Arduino Nano

- Arduino Uno R4 WiFi
- ELEGOO UNO R3 boards
- ELEGOO Super Starter Kit and UNO R3 Project Most Complete Starter Kit
- ELEGOO UNO R3 2.8 in. TFT Touch Screen with SD Card Socket
- ELEGOO UNO R3 Smart Robot Car Kit V4 with Camera
- Raspberry Pi 4 Model B (4 GB)
- Raspberry Pi Pico 2 W with soldered headers
- Teensy 4.1 with pins
- ESP-WROOM-32 / ESP-32S development boards
- ESP32-CAM and ESP32-CAM-MB camera modules
- Nordic nRF52840-DK (Bluetooth 5.x LE, Thread, Zigbee, ANT)

Storage and Computer Peripherals

- microSD cards (32–256 GB) with SD adapters
- USB-C SD card readers
- M.2 NVMe solid-state drive (512 GB)
- Powered 7-port USB 3.0 hubs
- Wired USB keyboards and mice
- Mechanical keyboard switches (MX 5-pin), arcade pushbuttons, joysticks, 3×4 keypads, and 2-player arcade control kits
- Infrared remote controls (Elegoo)
- Cat6 Ethernet cable

Sensors: Motion, Position, and Ranging

- PIR motion sensors (HC-SR501)
- Micro limit switches
- Thin-film force-sensitive resistors (500 g and 20 g–6 kg)
- 6-axis IMUs (MPU-6050) and 9-axis IMUs (MPU-9250)
- BNO055 9-DOF absolute orientation fusion breakouts
- ADXL345 and MMA8451 3-axis accelerometers
- Photoresistors / light-dependent resistors (5 mm, 5539)
- HC-SR04 ultrasonic and JSN-SR04T waterproof ultrasonic distance sensors
- VL53L0X and VL53L1X infrared time-of-flight sensors (Class 1 emitters)
- 8×8 matrix ToF depth-imaging sensor (VL53L7CX, 64-pixel)
- Slamtec RPLIDAR C1 360° laser scanner (Class 1 laser product) (Restricted)
- NEO-6M GPS/GNSS positioning modules with antennas
- Strain-gauge load cells (5 kg) with HX711 amplifier breakouts
- AT42QT1011 capacitive touch breakouts

- APDS-9930 proximity and ambient light sensors
- ADS1115 16-bit ADC modules

Sensors: Environmental and Chemical

- Temperature and humidity sensor modules
- Capacitive soil moisture sensors and water level / leak sensors
- DS18B20 waterproof temperature probes and TMP117 high-precision temperature sensors (Qwiic)
- BME280 pressure, temperature, and humidity sensors
- Sensirion SEN54 particulate matter, VOC, humidity, and temperature sensor
- Analog pH sensor kits (Restricted)
- Analog TDS / water-quality conductivity probes (Restricted)

Camera and Imaging

- Raspberry Pi Camera Module 3 (12 MP IMX708, autofocus)
- Raspberry Pi High Quality Camera (12.3 MP IMX477)
- Arducam mini camera shields (OV2640) for Arduino and Raspberry Pi Pico
- ESP32-CAM OV2640 camera modules
- USB webcams
- DFRobot HuskyLens AI vision sensor
- MLX90640 32×24 infrared thermal imaging camera

Audio

- INMP441 I2S MEMS microphone modules
- USB mini microphones
- Sound sensor modules
- ReSpeaker Mic Array v2.0 and ReSpeaker XVF3800 USB Mic Array
- USB conference speakerphone
- MAX98357A I2S class-D audio amplifier breakouts
- Small speakers (3 W, 4 Ω and 8 Ω) and USB mini speakers
- Active and passive piezo buzzers, piezo disc transducers, and piezo pickups

Biosignal and Physiological Sensing (Restricted)

All items in this category are non-medical research and hobbyist devices used for signal acquisition only. None of them deliver current to the body, and no stimulation devices (for example TENS units) are provided, permitted, or purchased. Use is governed by the "Biosignal and Wearable Device Protocol" section below.

- NeuroSky MindWave Mobile 2 EEG headset

- Gold cup EEG electrodes with DIN 1.5 mm plugs
- MyoWare 2.0 cable shields and link shields
- Electrode lead wire adapters (2.5 mm plug to 3.5 mm snaps)
- MAX30102 pulse oximetry and heart-rate sensors, and photoplethysmography pulse sensors
- Single-use disposable EKG/ECG electrodes and self-adhesive electrode pads

Wearable and Head-Mounted Devices (Restricted)

- Meta Ray-Ban AI glasses (Gen 2)

Haptics

- Coin / ERM vibration motors
- DRV2605L haptic motor driver breakouts

Robotics and Motors

- SG90-class micro servos and MG995 / MG996R metal-gear servos
- Geared DC motors with gearbox (12 V, 100 RPM, 37 mm)
- Bipolar stepper motors (Nema 17, 26 Ncm)
- TB6612FNG dual motor drivers, L298N dual H-bridge drivers, A4988 stepper drivers, and PCA9685 16-channel PWM servo driver shields
- 5 V relay modules
- Mini pan/tilt kits with micro servos
- 3-pin servo extension cables
- Propellers
- 4WD omnidirectional mecanum robot chassis kits (Restricted — floor operation in designated zone only)

Connectivity and Identification

- Bluetooth 5.x LE, Thread, Zigbee, and ANT via Nordic nRF52840-DK
- Wi-Fi and Bluetooth via ESP32 and Arduino Uno R4 WiFi boards
- RC522 RFID reader/writer module

Prototyping, Passives, and Interconnect

- Solderless breadboards, full-size and mini
- Male-male, male-female, and female-female jumper wire kits and loose jumper wires
- Resistor kits (1/4 W, assorted values) and individual resistors
- Electrolytic capacitor kits (0.1 μ F–1000 μ F)
- LED assortments (3 mm and 5 mm), WS2812B addressable rings and strips
- Infrared emitter and receiver diode pairs (940 nm, 5 mm)

- Break-away header pin strips
- Heat shrink tubing kits

Other Components

- Ultrasonic fogger / mist maker for demonstration use (Restricted — staff-supervised, on a tray, away from mains outlets)